

Secondary Storage Systems



John Jose

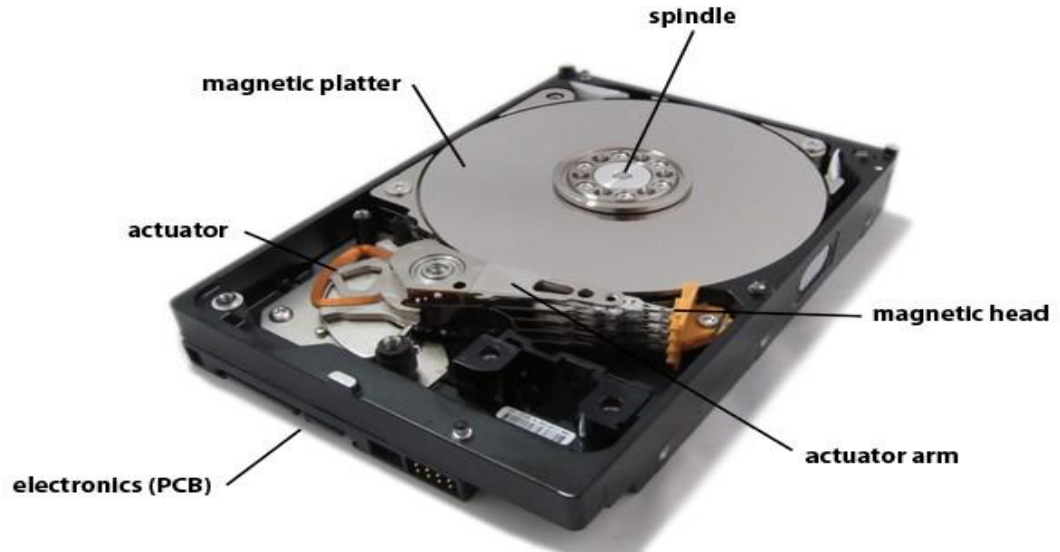
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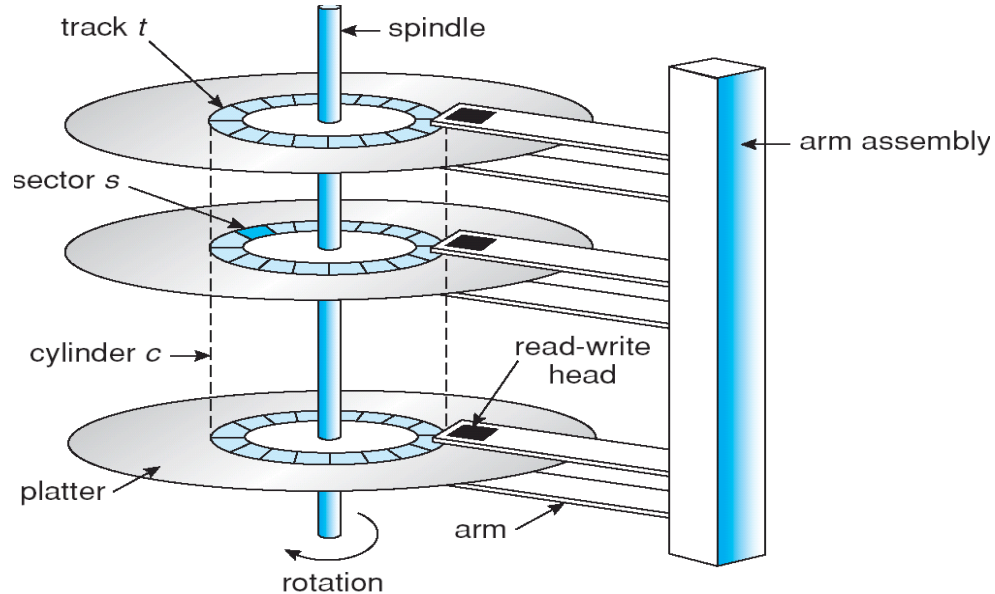
Mass Storage - Hard Disk Drive

- ❖ Systems today need to store many terabytes of data.
 - ❖ Primary level of permanent storage is hard disk.
 - ❖ Electromechanical
 - ❖ Rotating disks
 - ❖ Arm assembly
 - ❖ Electronics
 - ❖ Disk controller
 - ❖ Cache
 - ❖ Interface controller
-
- The image shows a top-down view of an open hard disk drive. The central spindle is visible, with a magnetic platter mounted on it. An actuator arm is positioned over the platter. The electronics (PCB) are located at the bottom of the drive. Labels with leader lines identify these components: spindle, magnetic platter, actuator, and electronics (PCB).



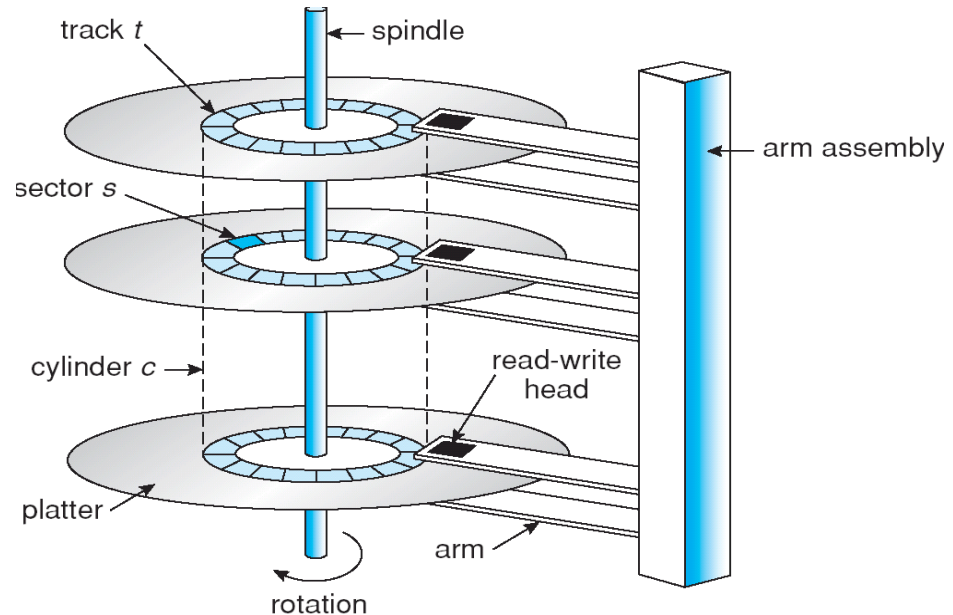
Hard Disk Drive Organization

- ❖ Hard disk drive consists of spinning disks with heads that move over the disks and store data in tracks and sectors.
- ❖ The heads read and write data in concentric rings called tracks.
- ❖ Tracks are divided into sectors, which normally store 512 bytes each.



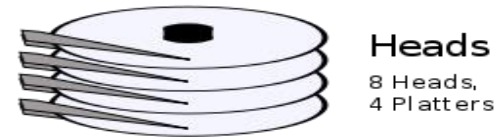
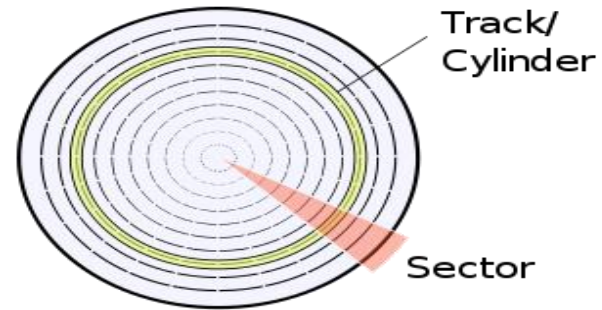
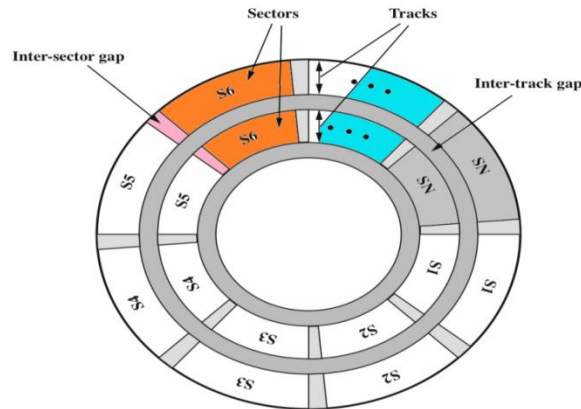
Hard Disk Drive Organization

- ❖ Platter diameters: 3.7", 3.3", 2.6"
- ❖ RPMs: 5400, 7200, 10000, 15000 [0.5 to 1%variation]
- ❖ Number of platters: 1-5
- ❖ Read/write head



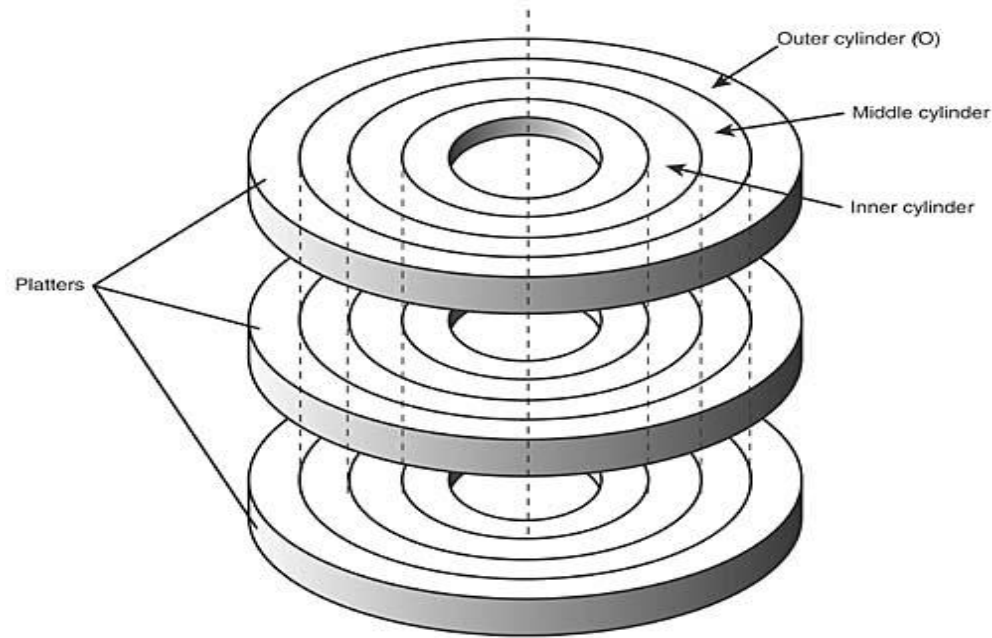
Hard Disk Drive Operation

- ❖ One side of a platter is called a head.
- ❖ HDD can have multiple platters, depending on their design and storage capacity.
- ❖ On the heads, there are concentric rings (tracks) and pieces of rings (sectors)



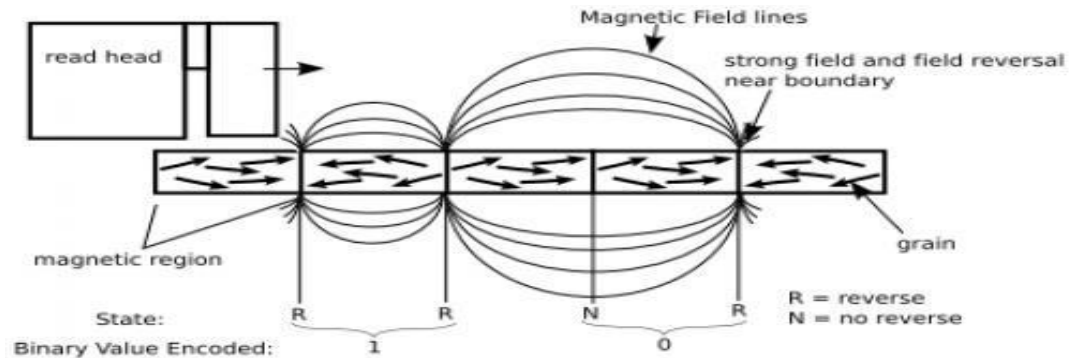
Hard Disk Drive Organization

- ❖ Cylinder: 3D collection of track 'n' of each surface of all platters.

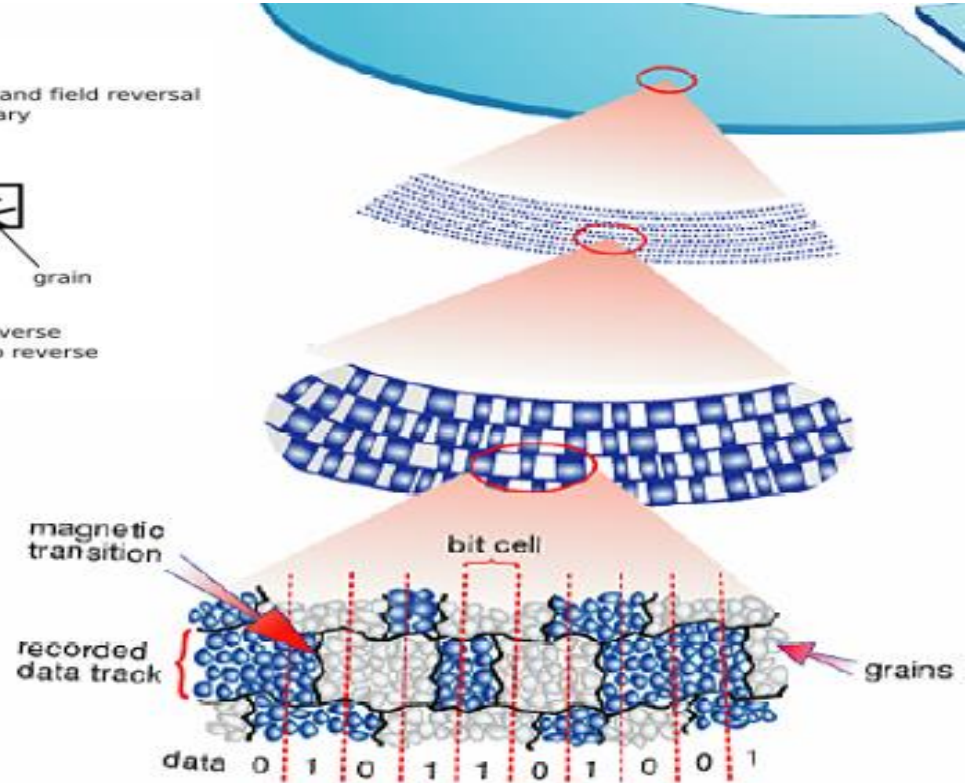
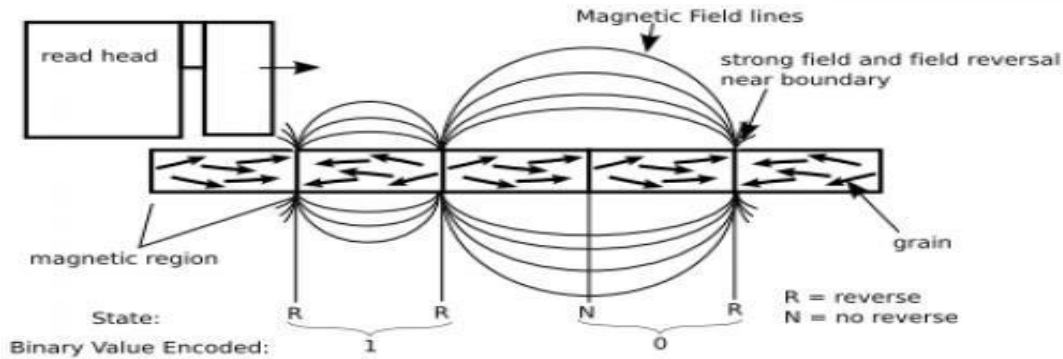


Logic storage in Hard Disk Drive

- ❖ Bit-cell composed of magnetic grains : 50-100 grains/bit
- ❖ Size of grains is order of 10 nm.
- ❖ **'0'** Region of grains of uniform magnetic polarity
- ❖ **'1'** Boundary between regions of opposite magnetization
- ❖ The read-and-write head is used to detect and modify the magnetization of the material immediately under it.

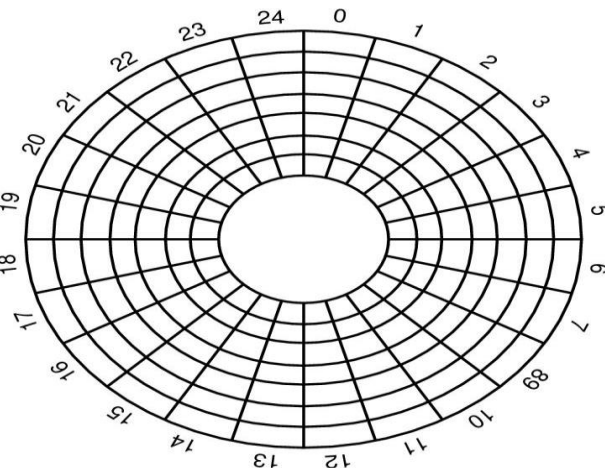
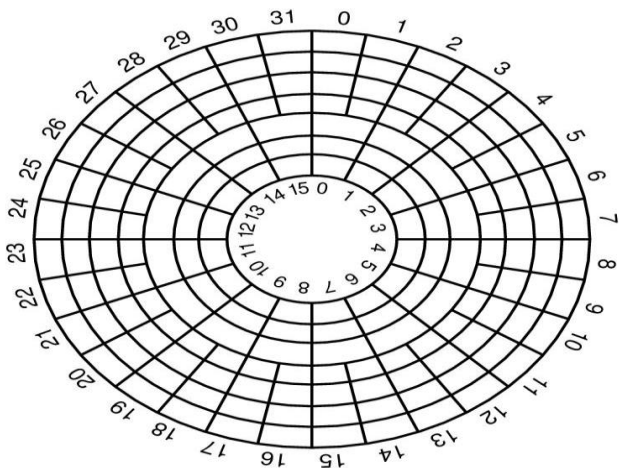


Logic storage in Hard Disk Drive



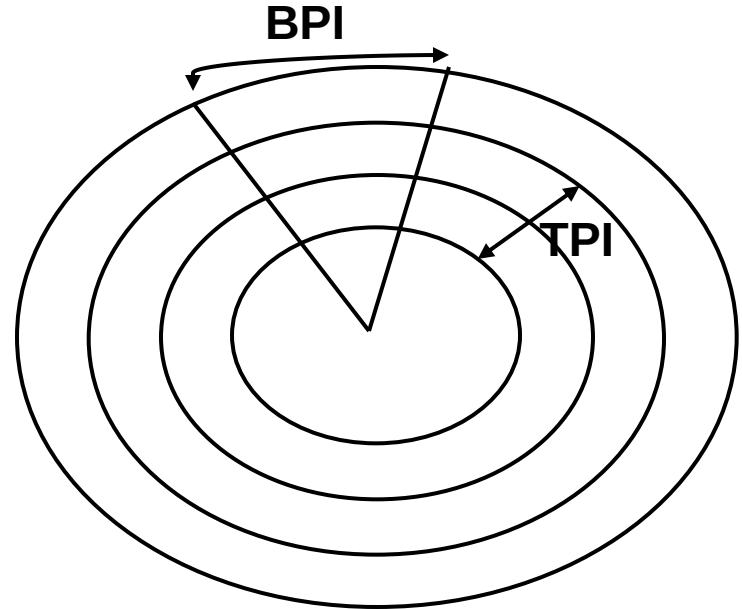
HDD- Bit Density

- ❖ Reduce bit density per track for outer layers. Constant Linear Velocity.
- ❖ Have more sectors per track on the outer layers, and increase rotational speed when reading from outer tracks. Constant Angular Velocity.

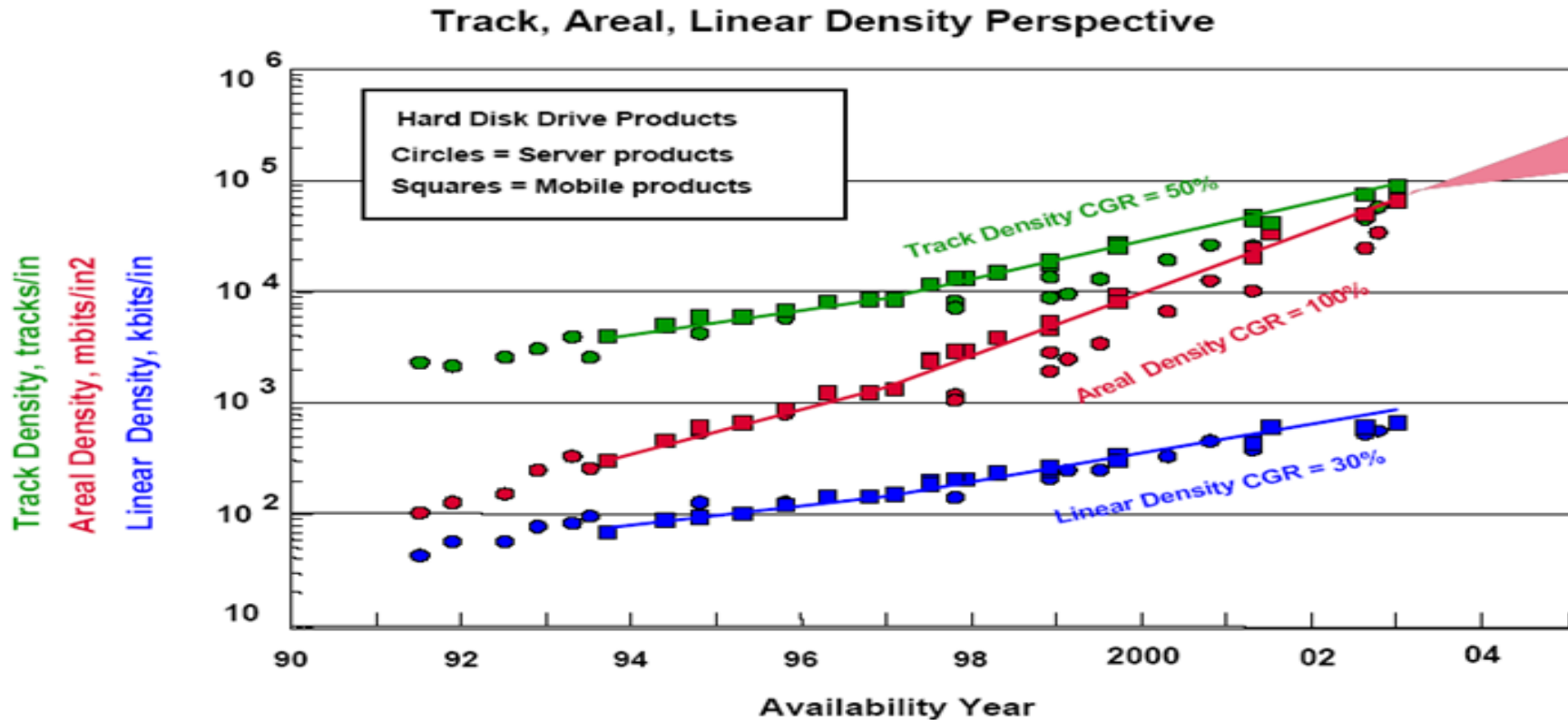


Storage Density

- ❖ Determines both capacity and performance
- ❖ Density Metrics
 - ❖ Linear density (Bits/inch or BPI)
 - ❖ Track density (Tracks/inch or TPI)
 - ❖ Areal Density = $\text{BPI} \times \text{TPI}$

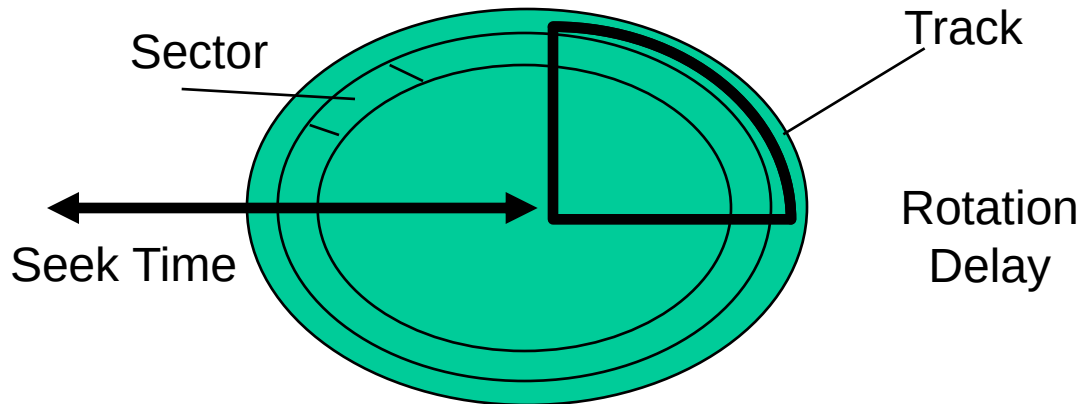


Storage Density Trends



Disk Access Time

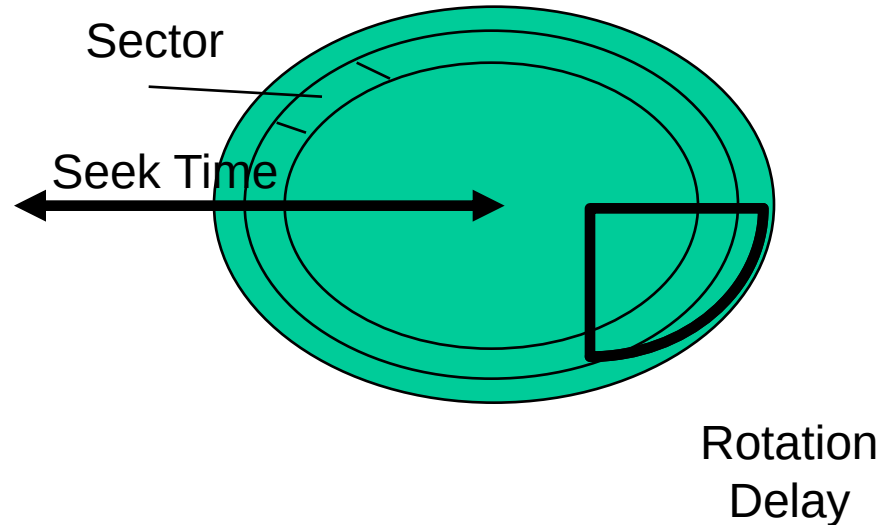
- ❖ To read from disk, we must specify:
 - ❖ cylinder #, surface #, sector #, size, memory address
- ❖ Transfer time includes:
 - ❖ Seek time: to get to the track
 - ❖ Rotational Latency: to get to the sector and
 - ❖ Transfer time: get bits off the disk



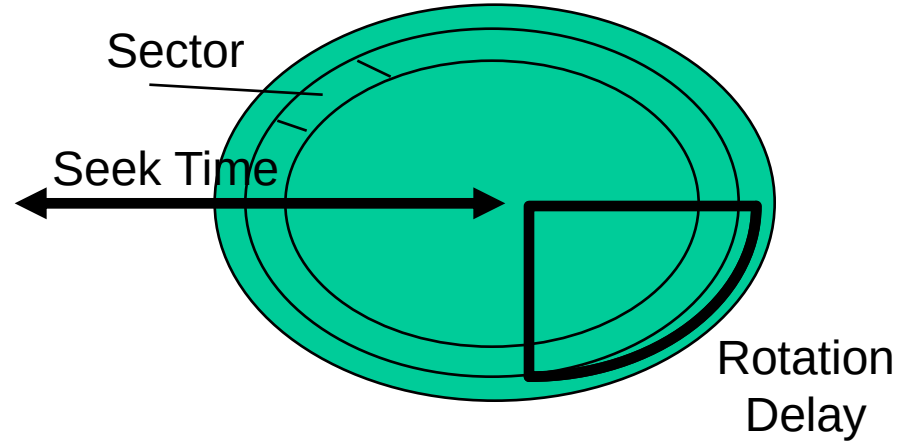
Seek Time

❖ Seek time depends on:

- ❖ Inertial power of the arm actuator motor
- ❖ Distance between outer-disk recording radius and inner-disk recording radius (data-band)
- ❖ Depends on platter-size

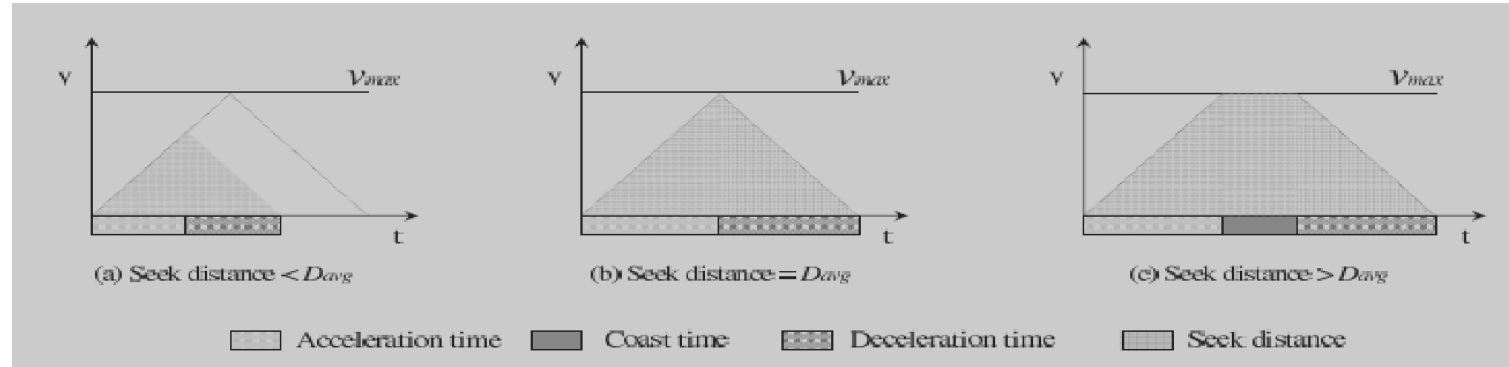


Seek Time



- ❖ **Components of a seek:**
- ❖ Speedup: Arm accelerates
- ❖ Coast: Arm moving at maximum velocity (long seeks)
- ❖ Slowdown: Arm brought to rest near desired track
- ❖ Settle: Head is adjusted to reach the access the desired location

Variations in Seek Time



- ❖ Very short seeks (2-4 cylinders)
 - ❖ Settle-time dominates
- ❖ Short seeks (100-200 cylinders)
 - ❖ Speedup/Slowdown-time dominates
- ❖ Longer seeks (> 200 cylinders)
 - ❖ Coast-time dominates
- ❖ With smaller platter-sizes and higher TPI
 - ❖ Settle-time becoming more important



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